

Zeal Education Society's
Zeal College of Engineering & Research, Pune
Department of Electronics and Computer Engg

Subject : Basic Electronics Engg

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Unit V : Electronic Measurement and Instrumentation

- Electronics measurements:
 - Frequency measurements and conversions in various units like Hz, KHz, MHz etc,
 - Voltage, current and power Measurement units
 - Measurement units for resistance, conductance, impedance, capacitance and inductance.
- Electronic Instruments:
 - Principles and block diagram of
 - Digital multimeter
 - Function Generator
 - Digital Storage Oscilloscope (DSO)
 - Power scope
 - AC/DC power supply
 - Auto transformer
 - Analog ammeter and voltmeter

1. Frequency Measurements

- Frequency measures based on how many cycles occur per second.
- The base unit is the **Hertz (Hz)**

Common Units & Conversions

- In electronics, frequency is usually increased or decreased by factors of 1000

Unit	Symbol	Value in Hz	Scientific Notation
Hertz	Hz	1 Hz	10^0
Kilohertz	kHz	1,000 Hz	10^3
Megahertz	MHz	1,000,000 Hz	10^6
Gigahertz	GHz	1,000,000,000 Hz	10^9

Conversion Rule:

- **To a larger unit:** Divide by 1,000 (e.g., 5000 Hz = 5 kHz)
- **To a smaller unit:** Multiply by 1,000 (e.g., 2MHz = 2,000 kHz).

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2. Voltage, Current, and Power

- These are the three "pillars" of any electrical circuit, related by Ohm's Law and the Power Law.

Parameter	Unit	Symbol	Base Formula
Voltage	Volt	V	$V = I R$
Current	Ampere	A	$I = V / R$
Power	Watt	W	$P = V I$

- **Voltage** is the Potential Difference
 - Common units: **mV** (millivolt), **V** (Volts) and **kV** (kilovolt)
- **Current** is the "flow" of charge.
 - Common units: **A** (ampere), **mA** (milliampere) and **μA** (microampere)
- **Power** is the "work rate."
 - Common units: **mW** (milliwatt), **W**(watt) and **kW** (kilowatt).

3. Passive Components & Properties

- These units measure how a component interacts with electrical energy.

Resistance, Conductance, and Impedance

- These units all deal with how much a circuit "opposes" or "allows" current.
- **Resistance (R):**
 - Measured in **Ohms**
 - It is the opposition to DC flow.
- **Conductance (G):**
 - Measured in **Siemens (S)** (formerly *mhos*).
 - It is the reciprocal of resistance ($G = 1/R$)
- **Impedance (Z):**
 - Measured in **Ohms**
 - It is the total opposition to AC flow, combining resistance and reactance.

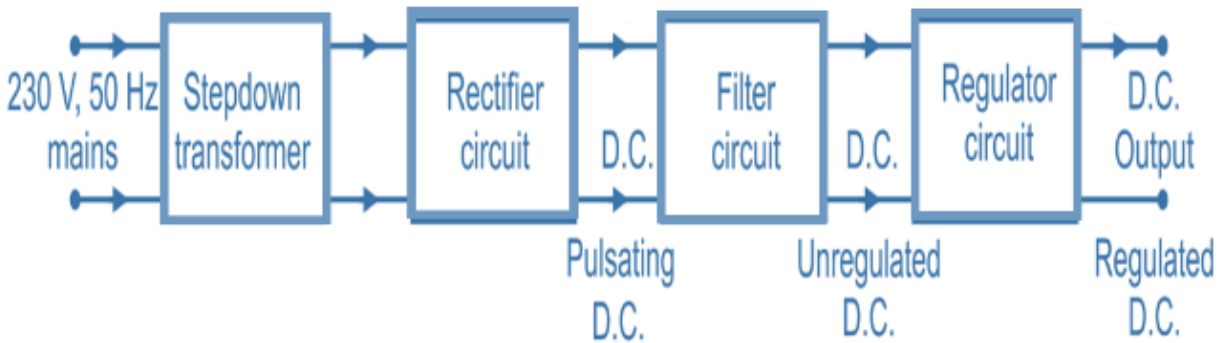
Capacitance and Inductance

These measure the ability to store energy in electric or magnetic fields.

Parameter	Unit	Symbol	Typical Prefixes used in Electronics
Capacitance (C)	Farad	F	μF, nF and pF
Inductance (L)	Henry	H	mH, μH

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Block Diagram of AC/DC Power Supply



- A regulated power supply is an electronic circuit that converts unregulated AC or DC input voltage into a regulated output voltage.
- It is an essential component in many electronic devices, providing a consistent and reliable source of power.
- Figure shows a block diagram of a regulated power supply

1. Step Down Transformer:

- The mains AC voltage is connected to input of Step Down Transformer which step down the AC voltage to the required level.

2. Rectifier Circuit:

- The step down AC voltage is applied to Rectifier circuit.
- Rectifier circuit converts AC voltage into Pulsating DC voltage.
- A pulsating DC output voltage consists of ripples.
- These ripples are removed using filter circuit.

3. Filter Circuit:

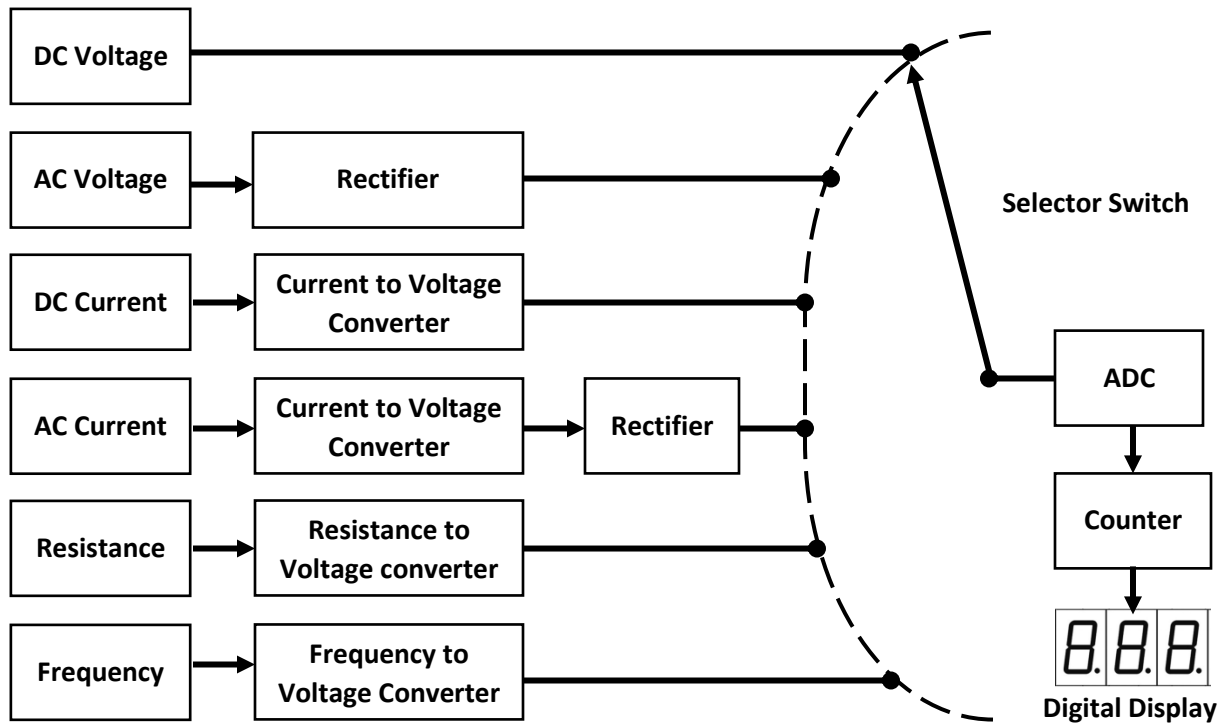
- The ripples present in the output of Rectifier circuit are removed using Filter circuit.
- The filter circuit gives unregulated DC voltage.

4. Regulator Circuit:

- The output of filter circuit is given to regulator circuit.
- It gives a constant DC output voltage irrespective of change in input voltage.

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Block Diagram of Digital Multimeter



- A *digital multimeter* (DMM) is a measuring instrument used to measure various electrical quantities.
- The standard measurements that are performed by a DMM are current, voltage and resistance.
- Apart from these, a digital multimeter can also measure temperature, frequency, capacitance, continuity, transistor gains etc.

DC Voltage Mode:

- The applied input DC voltage is given to analog to digital converter (ADC) through selector switch.
- It is used to measure DC Voltage.

AC Voltage Mode:

- The applied input AC voltage is fed to a full-wave rectifier.
- The resulting DC is fed to analog to digital converter (ADC) and finally to the display system.
- It is used to measure AC Voltage.

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DC Current Mode :

- The input DC current is converted into corresponding DC voltage using current to voltage converter.
- The resulting DC is fed to analog to digital converter (ADC) and finally to the display system.
- It is used to measure DC Current.
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In AC Current Mode :

- The input AC current is converted into corresponding AC voltage using current to voltage converter.
- After AC to DC conversion using rectifier, the resulting DC voltage is fed to analog to digital converter (ADC) and finally to the display system.
- It is used to measure AC Current.

Resistance Mode :

- The resistance to be measured is converted into corresponding DC voltage using resistance to voltage converter.
- The resulting DC is fed to analog to digital converter (ADC) and finally to the display system.
- It is used to measure resistance.

Frequency Mode :

- The frequency to be measured is converted into corresponding DC voltage using frequency to voltage converter.
- The resulting DC is fed to analog to digital converter (ADC) and finally to the display system.
- It is used to measure frequency.

Selector Switch :

- There is a rotary switch to select the types of measurement to be made and range that is needed.

Analog-to-Digital Converter (ADC):

- Allows is used to convert analog input into digital values that can be processed for counting

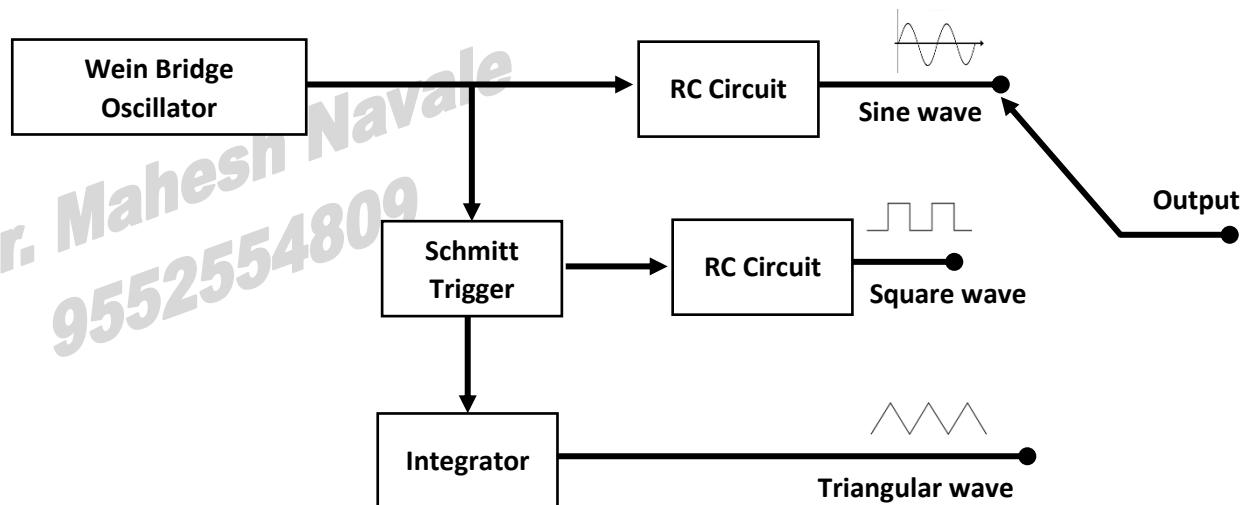
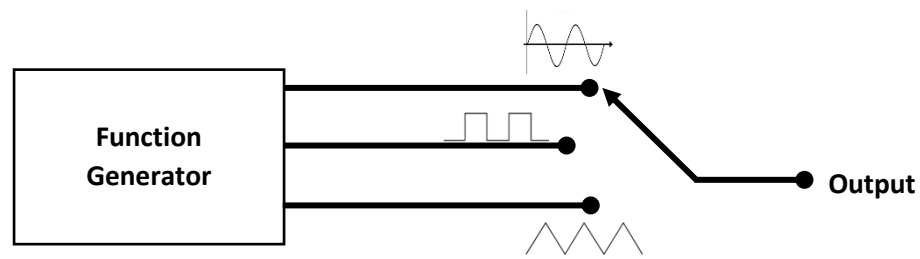
Counter:

- It is used to count the pulses and sent to the display device to display the final reading.

Digital Display:

- It is used to display the final reading of the parameter to measure.

Block Diagram of Function Generator



- The function generator also called as signal generator.
- It generates sine wave, square wave and triangular wave.

Sine Wave

- The sine wave generating circuit such as wein bridge circuit will produce a sine wave output.
- The output frequency can be varied using RC circuit

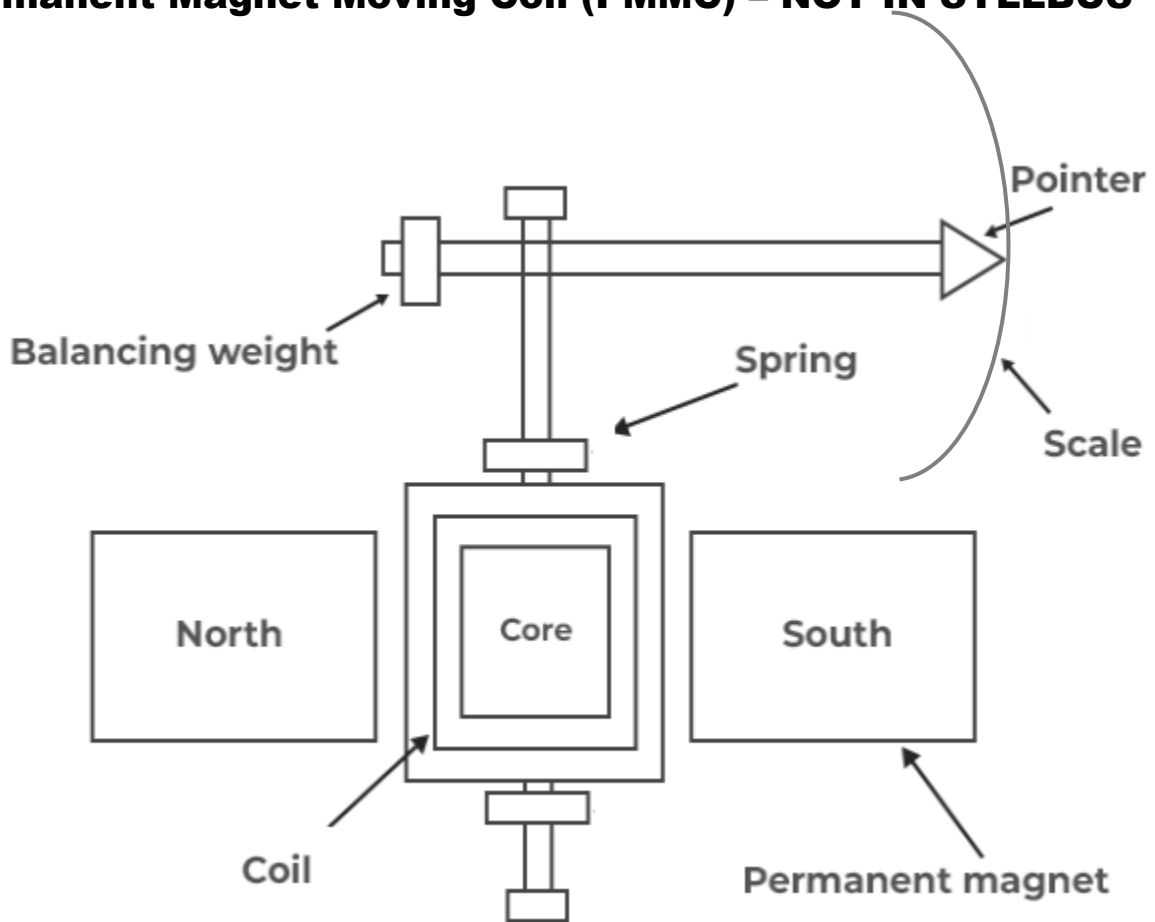
Sine Wave

- The sine wave output generated by wein bridge circuit is applied to Schmitt trigger circuit.
- Schmitt trigger circuit is a circuit which converts applied sine wave into square wave.
- The output frequency can be varied using RC circuit

Triangular Wave

- The square wave generated by Schmitt trigger circuit is applied to an RC integrator circuit.
- An integrator is used to convert square wave into respective triangular wave.

Permanent Magnet Moving Coil (PMMC) – NOT IN SYLLBUS



A permanent magnet moving coil instrument can measure DC voltage & current.

- The working principle of these instruments is the same as that of the galvanometer, the difference being that a direct reading is provided with a pointer & a scale.
- It has a permanent magnet & a rectangular coil having number of turns wound on former.
- An iron core is provided inside the former to provide an easy path for the magnetic lines of force.
- The coil is mounted on a spindle.
- The springs are used to supply current to the moving coil.
- The pointer is provided on the spindle to indicate the magnitude of quantity being measured.
- When operating current flows through the coil, it produces a magnetic flux. Direction of magnetic flux depends upon the direction of current flowing through the coil.
- Thus the direction of force acting on a moving coil & hence direction of movement of coil depends upon the direction of current through the coil.

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- If the direction of DC current flow through the coil is reversed, the direction of deflecting torque & movement of pointer will also be reversed.
- The Magnitude of the force (F) experienced by both sides of the coil and it is given by,

$$F = BIL$$

where,

B – flux density

L – length of the coil

I – current flowing through coil

Advantages of PMMC Instrument

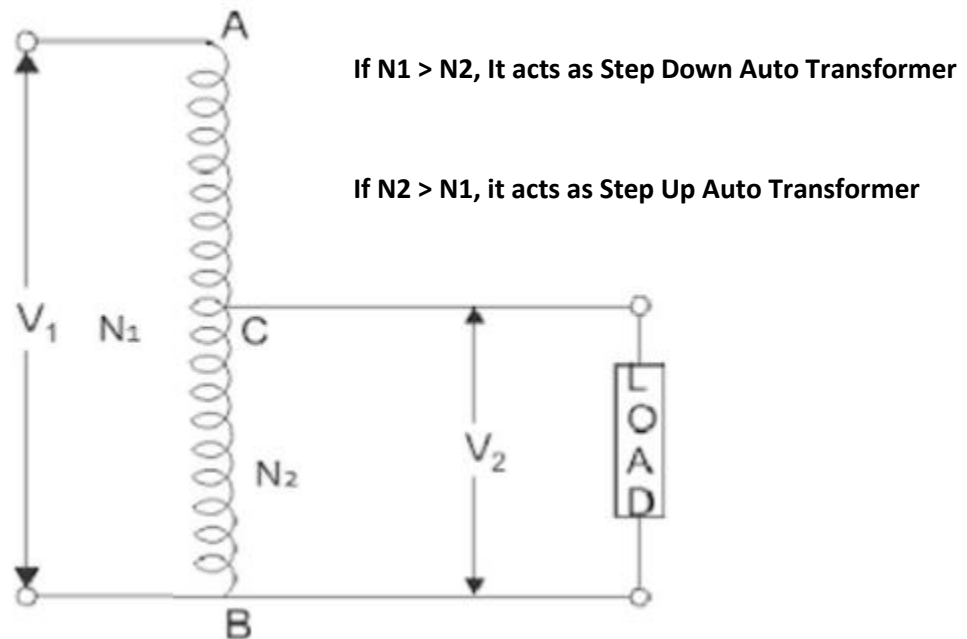
- Scale of these instruments is uniform.
- These are accurate & reliable.
- It's power consumption is low.

Disadvantages of PMMC Instrument

- These instruments can operate DC supply only.
- These are costlier than moving iron instrument.

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Autotransformer



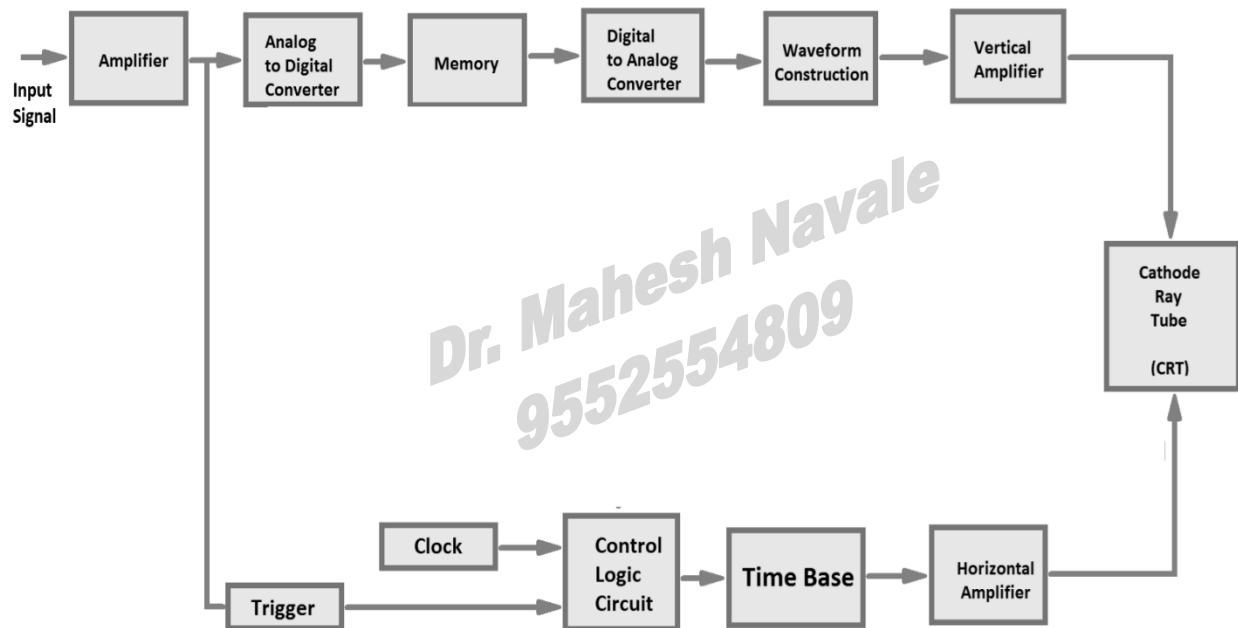
- An autotransformer is a type of electrical transformer with only one winding
- In an ordinary transformer, the primary & secondary windings are electrically insulated from each other
- Whereas in an autotransformer, one single winding is used as primary winding as well as secondary winding.
- The primary winding AB having N_1 turns & BC is secondary winding having N_2 turns.
- The tapping is taken at point C.
- The tapping may be fixed or variable.
- The working of autotransformer is based on the principle of electromagnetic induction similar to the two winding.
- When AC voltage V_1 is applied across AB, an alternating flux is setup in the core, it induces an e.m.f. E_1 in the winding AB.
- The part of the e.m.f. is taken in the secondary circuit.

Let, V_1 = Primary applied voltage
 V_2 = Secondary load voltage
 N_1 = Primary current
 N_2 = Secondary current

Transformation Ratio is given by $\frac{V_2}{V_1} = \frac{N_2}{N_1}$

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Digital Storage Oscilloscope (DSO)

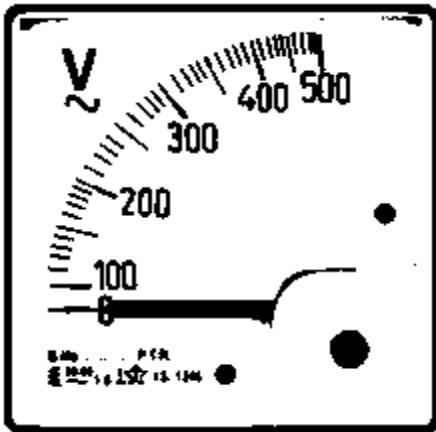


- The Digital Storage Oscilloscope or DSO is a device that helps to see the waveforms as well as save the waveforms in a digital format in its digital storage unit or memory.
- The digital storage oscilloscope takes the input signal in the analog format and stores it in its memory.
- After that, it takes the data from its memory to display.
- Generally, it stores the signal or data in a binary format such as 0 and 1.
- The DSO involves in waveform displaying, digital storage, signal processing, etc.
- The main feature of the DSO is it provides the function to analyze the signals or waveforms later as stored in its memory using the Digital Signal Processing
- The main parts of the DSO are,
 - A/D and D/A converter
 - Memory
 - Logic Control Circuit
 - Amplifier
 - Time Base Circuit
 - Cathode Ray Tube
- DSO accepts the signal in the form of an analog signal.
- An analog signal is amplified using amplifier.
- After amplifying the analog to the digital conversion process is carried out.

- After converting the analog signal into a digital signal, it stores the signal in its memory in the form of a digital signal or binary form such as 0 and 1. During the waveform display, it takes the data from the memory.
- Control Logic Circuit controls so many things. It takes the clock pulses and triggers pulses. It provides the output to the time base circuit.
- In a Digital Storage Oscilloscope (DSO), the Control Logic Circuit is responsible for managing the overall operation and control of the instrument.
- Digital to Analog Converter(D/A) is used to display waveforms or analyzes the signal. The digital data or signal is stored in the memory is being converted into an analog signal.
- Waveform Construction: The analog signal converted from the digital signal stored in the memory is constructed in the form of waveform through this waveform construction circuit.
- CRT(Cathode Ray Tube): The cathode ray tube display visualizes the waveforms on the screen. It has two plates Horizontal plate and a vertical plate. The horizontal plate is connected through the horizontal amplifier whereas the vertical plate is connected through the vertical amplifier.

Voltmeter :

- A **voltmeter**, also known as a voltage meter, is an instrument that measures the voltage or potential difference between two points of an electronic or electrical circuit.
- Usually, the voltmeter is used for Alternating Current (AC) circuits or Direct Current (DC) circuits..



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- A voltmeter measures voltages usually calibrated in volts, millivolts (0.001 volt), or kilovolts (1,000 volts).
- In order to measure a voltage, a **voltmeter is connected in parallel** to a device.
- It is connected in parallel with the circuit, mainly because the same voltage drop occurs across it.
- A voltmeter also has high internal resistance.

Voltmeter Symbol

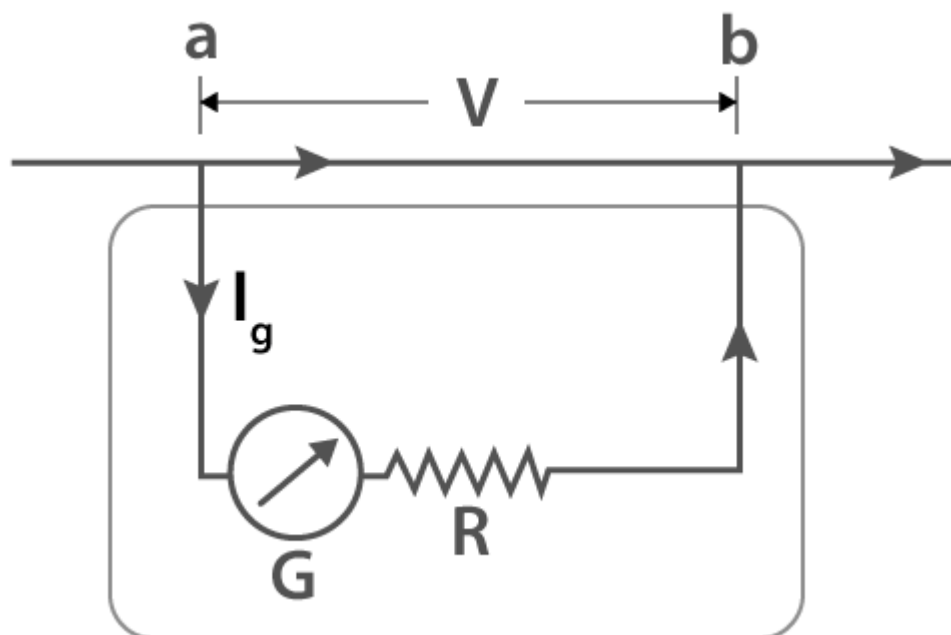
- The voltmeter is usually represented by the letter V, which is placed inside a circle adjoining two terminals.



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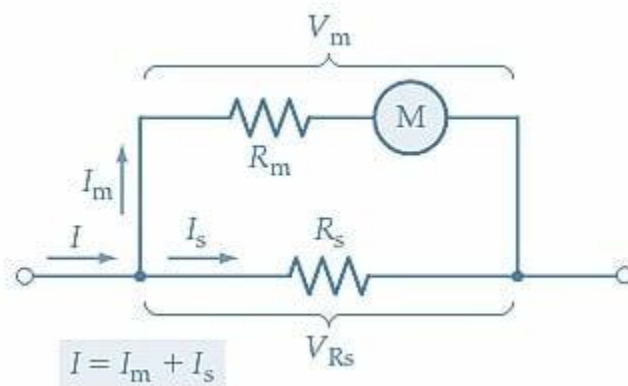
Analog Voltmeter

- An analogue voltmeter is used mainly for measuring the AC voltage.
- The reading is displayed with the help of a pointer that is fixed on the calibrated scale.
- The movement of the pointer is affected by the torque that is acting on it.
- The magnitude of the torque that is developed is directly proportional to the voltage being measured.
- A **galvanometer** (current meter) that is sensitive and is part of a high resistance series is what makes a basic analogue voltmeter.
- The meter should have high internal resistance, or else the circuit operation during the test would be interrupted by drawing current significantly.
- The voltage range displayed by the meter is determined by the series resistance value and the galvanometer sensitivity.



Analog Ammeter

- An analog current meter, commonly known as an ammeter, works by converting electrical current into mechanical motion to move a pointer across a calibrated scale.
- The most common mechanism used is the Permanent Magnet Moving Coil (PMMC) movement
- When current flows through a wire coil placed within the field of a permanent magnet, it creates its own magnetic field.
- The interaction between the coil's magnetic field and the permanent magnet's field generates a force (torque) that causes the coil to rotate.
- A needle or pointer is attached to this rotating coil.
- As the coil turns, the needle moves across a dial marked with numerical values.
- Control springs provide an opposing force.
- The needle stops moving when the electromagnetic force (from the current) is exactly balanced by the spring's tension.
- A shunt resistance is employed in connected with the PMMC (permanent magnet moving coil) to measure larger currents accurately.
- In the DC ammeter, a resistor known as a shunt is shown connected in parallel with the PMMC.
- Figure shows the equivalent circuit of the complete instrument, including the coil resistance R_m and the shunt resistance R_s .



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